

EXERCISE SHEET 2

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VECTORS AND GEOMETRY OF SPACE

- (1) Find the vector represented by the directed line segment with initial point $A(2, -3, 4)$ and terminal point $B(-4, 1, 1)$

$$\vec{AB} = \langle -4 - 2, 1 - (-3), 1 - 4 \rangle = \langle -6, 4, -3 \rangle$$

- (2) Find $|\vec{a}|$, $\vec{a} - \vec{b}$, $2\vec{a}$ and $3\vec{a} + 4\vec{b}$ for the following vectors

(a) $\vec{a} = \langle -4, 3 \rangle$ $\vec{b} = \langle 6, 2 \rangle$

(b) $\vec{a} = \vec{i} - 2\vec{j} + \vec{k}$ $\vec{b} = \vec{i} + 2\vec{k}$

(a) $|\vec{a}| = \sqrt{(-4)^2 + 3^2} = 5$

$$\vec{a} - \vec{b} = \langle -4, 3 \rangle - \langle 6, 2 \rangle = \langle -10, 1 \rangle$$

$$2\vec{a} = \langle -8, 6 \rangle$$

$$\begin{aligned} 3\vec{a} + 4\vec{b} &= 3\langle -4, 3 \rangle + 4\langle 6, 2 \rangle \\ &= \langle -12, 9 \rangle + \langle 24, 8 \rangle \\ &= \langle 12, 17 \rangle \end{aligned}$$

(b) $\vec{a} = \vec{i} - 2\vec{j} + \vec{k} = \langle 1, -2, 1 \rangle$

$$\vec{b} = \vec{i} + 2\vec{k} = \langle 1, 0, 2 \rangle$$

$$|\vec{a}| = \sqrt{1^2 + (-2)^2 + 1^2} = \sqrt{6}$$

$$\vec{a} - \vec{b} = \langle 1, -2, 1 \rangle - \langle 1, 0, 2 \rangle = \langle 0, -2, -1 \rangle$$

$$2\vec{a} = \langle 2, -4, 2 \rangle$$

$$\begin{aligned} 3\vec{a} + 4\vec{b} &= 3\langle 1, -2, 1 \rangle + 4\langle 1, 0, 2 \rangle \\ &= \langle 7, -6, 11 \rangle \end{aligned}$$

- (3) Find a unit vector that has the same direction as given

(a) $\langle 9, -5 \rangle = \vec{u}$

(b) $8\vec{i} - \vec{j} + 4\vec{k}$

(a) $\vec{v} = \frac{\vec{u}}{|\vec{u}|} = \frac{\langle 9, -5 \rangle}{\sqrt{81+25}} = \left\langle \frac{9}{\sqrt{106}}, -\frac{5}{\sqrt{106}} \right\rangle$

(1)

(b) $\vec{v} = \langle 8, -1, 4 \rangle$

$$|\vec{v}| = \frac{\vec{v}}{|\vec{v}|} = \frac{\langle 8, -1, 4 \rangle}{81}$$

$$\vec{w} = \left\langle \frac{8}{81}, -\frac{1}{81}, \frac{4}{81} \right\rangle$$

3 Find $\vec{a} \cdot \vec{b}$ for the following vectors.

a) $\vec{a} = \langle 5, 0, -2 \rangle$ $\vec{b} = \langle 3, -1, 10 \rangle$

b) $\vec{a} = \langle s, 2s, 3s \rangle$ $\vec{b} = \langle t, -t, 5t \rangle$

c) $|\vec{a}| = 12$ $|\vec{b}| = 15$ the angle between $\vec{a}$ and $\vec{b}$ is $\pi/6$

a) $\vec{a} \cdot \vec{b} = \langle 5, 0, -2 \rangle \cdot \langle 3, -1, 10 \rangle = 5 \cdot 3 + 0 \cdot (-1) + (-2) \cdot 10 = -5$

b) $\vec{a} \cdot \vec{b} = \langle s, 2s, 3s \rangle \cdot \langle t, -t, 5t \rangle = s \cdot t - 2st + 15st = 14st.$

c) $\vec{a} \cdot \vec{b} = |\vec{a}| \cdot |\vec{b}| \cdot \cos \theta = 12 \cdot 15 \cdot \cos \frac{\pi}{6} = 180 \cdot \frac{\sqrt{3}}{2} = 90\sqrt{3}.$

4 Find the angle between the vectors

a) $\vec{a} = \langle \sqrt{3}, 1 \rangle$ $\vec{b} = \langle 0, 5 \rangle$

b) $\vec{a} = \vec{i} + \vec{k}$ $\vec{b} = \vec{i} + 2\vec{j} - 3\vec{k}$

c) $\vec{a} = \langle 6, -3, 2 \rangle$ $\vec{b} = \langle 2, 1, -2 \rangle$

a) $\vec{a} \cdot \vec{b} = \langle \sqrt{3}, 1 \rangle \cdot \langle 0, 5 \rangle = 0 \cdot \sqrt{3} + 5 \cdot 1 = 5.$

$$|\vec{a}| = \sqrt{(\sqrt{3})^2 + 1} = 2$$

$$|\vec{b}| = \sqrt{0 + 25} = 5.$$

$$\vec{a} \cdot \vec{b} = |\vec{a}| \cdot |\vec{b}| \cdot \cos \theta$$

$$5 = 5 \cdot 2 \cos \theta$$

$$\cos \theta = 1/2 \quad \theta = \pi/3$$

b) $\vec{a} = \langle 0, 1, 1 \rangle$ $\vec{b} = \langle 1, 2, -3 \rangle$

$$\vec{a} \cdot \vec{b} = 0 \cdot 1 + 1 \cdot 2 + 1 \cdot (-3) = -1$$

$$|\vec{a}| = \sqrt{1 + 1} = \sqrt{2} \quad |\vec{b}| = \sqrt{1 + 4 + 9} = \sqrt{14}$$

$$\vec{a} \cdot \vec{b} = |\vec{a}| \cdot |\vec{b}| \cdot \cos \theta$$

$$\cos \theta = \frac{-1}{\sqrt{2} \cdot \sqrt{14}}$$

$$\theta \approx 101^\circ$$

$$\textcircled{c} \quad \vec{a} \cdot \vec{b} = \langle 6, -3, 2 \rangle \cdot \langle 2, 1, -2 \rangle = 12 - 3 - 4 = 5$$

$$|\vec{a}| = \sqrt{36 + 9 + 4} = \sqrt{49} = 7$$

$$|\vec{b}| = \sqrt{4 + 1 + 4} = 3$$

$$\vec{a} \cdot \vec{b} = |\vec{a}| \cdot |\vec{b}| \cdot \cos \theta$$

$$\cos \theta = \frac{5}{7 \cdot 3} = \frac{5}{21}$$

$$\theta \approx 76^\circ$$

$\textcircled{5}$ Determine whether the given vectors are orthogonal, parallel, or neither.

$$\textcircled{a} \quad \vec{a} = \langle -5, 3, 7 \rangle \quad \vec{b} = \langle 6, -8, 2 \rangle$$

$$\textcircled{b} \quad \vec{a} = \langle 4, 6 \rangle \quad \vec{b} = \langle -3, 2 \rangle$$

$$\textcircled{c} \quad \vec{a} = -\vec{i} + 2\vec{j} + 5\vec{k} \quad \vec{b} = 3\vec{i} + 4\vec{j} - \vec{k}$$

$$\textcircled{d} \quad \vec{a} = 2\vec{i} + 6\vec{j} - 4\vec{k} \quad \vec{b} = -3\vec{i} - 9\vec{j} + 6\vec{k}$$

$\textcircled{a} \quad \vec{a} \cdot \vec{b} = \langle -5, 3, 7 \rangle \cdot \langle 6, -8, 2 \rangle = -5 \cdot 6 - 8 \cdot 3 + 2 \cdot 7 = -54 + 14 = -40$
 Since $\vec{a} \cdot \vec{b} \neq 0$, they are not orthogonal. Also they are not multiples of each other. Therefore, not parallel.

$\textcircled{b} \quad \vec{a} \cdot \vec{b} = 4(-3) + 6 \cdot 2 = 0$ since $\vec{a} \cdot \vec{b} = 0$, vectors are orthogonal.

$$\textcircled{c} \quad \vec{a} = \langle -1, 2, 5 \rangle \quad \vec{b} = \langle 3, 4, -1 \rangle$$

$$\vec{a} \cdot \vec{b} = (-1) \cdot 3 + 4 \cdot 2 + 5(-1) = 0$$

$\vec{a}$ and $\vec{b}$ are orthogonal.

$$\textcircled{d} \quad \vec{a} = 2\vec{i} + 6\vec{j} - 4\vec{k} \quad \vec{b} = -3\vec{i} - 9\vec{j} + 6\vec{k}$$

Since $\vec{b} = -\frac{3}{2}\vec{a}$ They are parallel.

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(6) Find the scalar projections of $\vec{b}$ onto $\vec{a}$

(a) $\vec{a} = \langle 3, -4 \rangle$ $\vec{b} = \langle 5, 0 \rangle$ (b) $\vec{a} = 2\vec{i} - 3\vec{j} + \vec{k}$ $\vec{b} = \vec{i} + 6\vec{j} - 2\vec{k}$

(a) Scalar projection of b onto a is $\text{comp}_{\vec{a}} \vec{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|}$

$$|\vec{a}| = \sqrt{9+16} = 5 \quad \text{comp}_{\vec{a}} \vec{b} = \frac{15}{5} = 3$$

$$\vec{a} \cdot \vec{b} = 3 \cdot 5 - 4 \cdot 0 = 15$$

Projection of $\vec{b}$ onto $\vec{a}$ is $\text{proj}_{\vec{a}} \vec{b} = \text{comp}_{\vec{a}} \vec{b} \cdot \frac{\vec{a}}{|\vec{a}|}$

$$\text{proj}_{\vec{a}} \vec{b} = 3 \cdot \frac{\langle 3, -4 \rangle}{5} = \left\langle \frac{9}{5}, -\frac{12}{5} \right\rangle$$

(b) $\vec{a} = \langle 2, -3, 1 \rangle$ $\vec{b} = \langle 1, 6, -2 \rangle$

$$\text{comp}_{\vec{a}} \vec{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|} = \frac{2 \cdot 1 - 3 \cdot 6 - 2 \cdot 1}{\sqrt{4+9+1}} = -\frac{18}{\sqrt{14}}$$

$$\text{proj}_{\vec{a}} \vec{b} = -\frac{18}{\sqrt{14}} \cdot \frac{\vec{a}}{|\vec{a}|} = -\frac{18}{\sqrt{14}} \cdot \frac{\langle 2, -3, 1 \rangle}{\sqrt{14}} = -\frac{18}{14} \langle 2, -3, 1 \rangle$$

$$\text{proj}_{\vec{a}} \vec{b} = \left\langle -\frac{18}{7}, \frac{54}{14}, -\frac{18}{14} \right\rangle$$

(7) If $\vec{a} = \langle 3, 0, -1 \rangle$, find a $\vec{b}$ such that $\text{scalar}_{\vec{a}} \vec{b} = 2$

$$\text{comp}_{\vec{a}} \vec{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|} = \left\langle \frac{3}{\sqrt{10}}, 0, -\frac{1}{\sqrt{10}} \right\rangle \cdot \langle x, y, z \rangle = 2$$

$$\vec{b} = \langle x, y, z \rangle$$

$$\frac{3}{\sqrt{10}}x - \frac{1}{\sqrt{10}}z = 2$$

$$3x + 0y - z = 2\sqrt{10}$$

so, we can take

$$\langle x, y, z \rangle = \langle s, t, 3s - 2\sqrt{10} \rangle$$

$$\frac{\vec{a}}{|\vec{a}|} = \frac{\langle 3, 0, -1 \rangle}{\sqrt{9+1}} = \left\langle \frac{3}{\sqrt{10}}, 0, -\frac{1}{\sqrt{10}} \right\rangle$$

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(14)

(8) For any $\vec{a}, \vec{b}$, prove the following equality

$$|\vec{a} + \vec{b}|^2 + |\vec{a} - \vec{b}|^2 = 2|\vec{a}|^2 + 2|\vec{b}|^2$$

$$\begin{aligned} |\vec{a} + \vec{b}|^2 &= (\vec{a} + \vec{b}) \cdot (\vec{a} + \vec{b}) = \vec{a} \cdot \vec{a} + \vec{a} \cdot \vec{b} + \underbrace{\vec{b} \cdot \vec{a}}_{=\vec{a} \cdot \vec{b}} + \vec{b} \cdot \vec{b} \\ &= |\vec{a}|^2 + 2\vec{a} \cdot \vec{b} + |\vec{b}|^2 \end{aligned}$$

$$|\vec{a} - \vec{b}|^2 = (\vec{a} - \vec{b}) \cdot (\vec{a} - \vec{b}) = \underbrace{\vec{a} \cdot \vec{a}}_{|\vec{a}|^2} - 2\vec{a} \cdot \vec{b} + \underbrace{\vec{b} \cdot \vec{b}}_{|\vec{b}|^2}$$

$$\begin{aligned} |\vec{a} + \vec{b}|^2 + |\vec{a} - \vec{b}|^2 &= |\vec{a}|^2 + 2\vec{a} \cdot \vec{b} + |\vec{b}|^2 + |\vec{a}|^2 - 2\vec{a} \cdot \vec{b} + |\vec{b}|^2 \\ &= 2|\vec{a}|^2 + 2|\vec{b}|^2 \end{aligned}$$

(9) Find the cross product of $\vec{a} \times \vec{b}$ and verify that it is orthogonal to $\vec{a}$ and $\vec{b}$

(a) $\vec{a} = \vec{i} + e^t \vec{j} + e^{-t} \vec{k}$ $\vec{b} = 2\vec{i} + e^t \vec{j} - e^{-t} \vec{k}$

(b) $\vec{a} = \langle t, t^2, t^3 \rangle$ $\vec{b} = \langle 1, 2t, 3t^2 \rangle$

(c)
$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & e^t & e^{-t} \\ 2 & e^t & -e^{-t} \end{vmatrix} = \vec{i}(-e^t \cdot e^{-t} - e^t \cdot e^{-t}) - \vec{j}(-e^{-t} - 2e^{-t}) + \vec{k}(e^t - 2e^t)$$

$$\vec{a} \times \vec{b} = -2\vec{i} - 3e^{-t}\vec{j} - e^t\vec{k} = \langle -2, -3e^{-t}, -e^t \rangle$$

$$\vec{a} \cdot (\vec{a} \times \vec{b}) = 1 \cdot (-2) + (t^2 \cdot (-3e^{-t})) + (t^3 \cdot (-e^t)) = -2 - 3t^2 e^{-t} - t^3 e^t$$

$$\vec{b} \cdot (\vec{a} \times \vec{b}) = 2 \cdot (-2) + e^t(-3e^{-t}) + (-e^{-t})(-e^t)$$

$$= -4 - 3 + 1 = -6$$

$$\textcircled{b} \quad \vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ t & t^2 & t^3 \\ 1 & 2t & 3t^2 \end{vmatrix} = \vec{i} (3t^4 - 2t^4) - \vec{j} (3t^3 - t^3) + \vec{k} (2t^2 - t^2)$$

$$\vec{a} \times \vec{b} = t^4 \vec{i} + 2t^3 \vec{j} + t^2 \vec{k} = \langle t^4, 2t^3, t^2 \rangle$$

$$\vec{a} \cdot (\vec{a} \times \vec{b}) = \langle t, t^2, t^3 \rangle \cdot \langle t^4, -2t^3, t^2 \rangle \\ = t^5 - 2t^5 + t^5 = 0$$

$$\vec{b} \cdot (\vec{a} \times \vec{b}) = \langle 1, 2t, 3t^2 \rangle \cdot \langle t^4, -2t^3, t^2 \rangle \\ = t^4 - 4t^4 + 3t^4 = 0$$

$\textcircled{10}$ Find two unit vectors orthogonal to both.

$$\vec{i} + \vec{j} + \vec{k} \quad \text{and} \quad 2\vec{i} + \vec{k}$$

$$\vec{a} = \vec{i} + \vec{j} + \vec{k} = \langle 1, 1, 1 \rangle \quad \vec{b} = 2\vec{i} + \vec{k} = \langle 2, 0, 1 \rangle$$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 1 & 1 \\ 2 & 0 & 1 \end{vmatrix} = \vec{i} (1) - \vec{j} (-2+1) + \vec{k} (-2) \\ = \vec{i} + \vec{j} - 2\vec{k} = \langle 1, 1, -2 \rangle$$

$(\vec{a} \times \vec{b})$ and also $-(\vec{a} \times \vec{b})$ are orthogonal to both $\vec{a}$ and $\vec{b}$. To find unit vectors, we will find the following

$$\vec{u} = \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|} = \frac{\langle 1, 1, -2 \rangle}{\sqrt{1+1+4}} = \left\langle \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, -\frac{2}{\sqrt{6}} \right\rangle$$

$$\vec{v} = -\frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|} = -\frac{\langle 1, 1, -2 \rangle}{\sqrt{6}} = \left\langle -\frac{1}{\sqrt{6}}, -\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}} \right\rangle$$

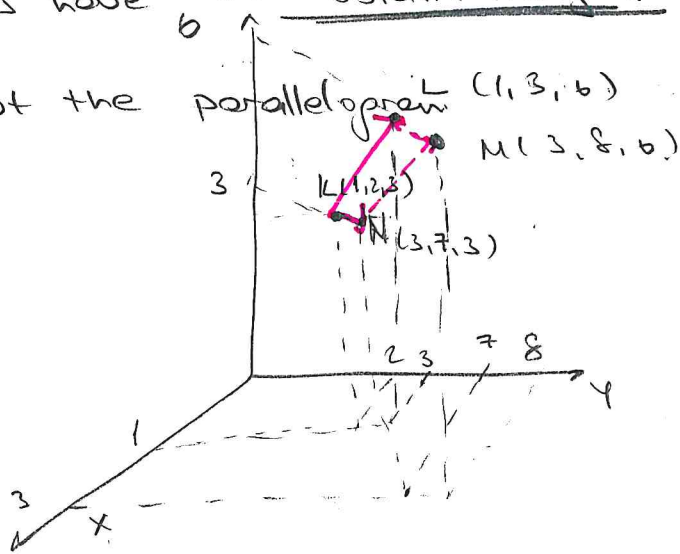
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11 Find the area of the parallelogram with vertices $K=(1,2,3)$

$L=(1,3,6)$ $M=(3,8,6)$ and $N=(3,7,3)$

You have to obtain the graph

Plot the parallelogram



As can be seen from the graph

The parallelogram KLMN is

$$|\vec{KL} \times \vec{KN}| = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 1 & 3 \\ 2 & 5 & 0 \end{vmatrix}$$

$$= \vec{i}(-15) - \vec{j}(-6) + \vec{k}(-2)$$

$$= \sqrt{(15)^2 + (6)^2 + (4)^2} = \sqrt{265}$$

Similarly you can also take $\vec{ML} \times \vec{MN}$

$$\vec{ML} = \langle 0, -1, -3 \rangle$$

$$\vec{MN} = \langle -2, -5, 0 \rangle$$

$$|\vec{ML} \times \vec{MN}| = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & -1 & -3 \\ -2 & -5 & 0 \end{vmatrix}$$

$$= \vec{i}(-15) - \vec{j}(-6) + \vec{k}(-2) = \sqrt{265}$$

(Properties of Determinant for any $[A]_{n \times n}$ $| -A | = (-1)^n |A|$)

For our case $\vec{ML} = -\vec{KL}$ so

$$\vec{KN} = -\vec{MN}$$

$$|\vec{KL} \times \vec{MN}| = (-1)^2 |\vec{ML} \times \vec{MN}|$$

12 Find the volume of the parallelepiped with edges

$\vec{P}\vec{Q}$, $\vec{P}\vec{R}$ and $\vec{P}\vec{S}$ where $P = (2, 0, -1)$ $Q = (4, 1, 0)$ $R = (3, -1, 1)$

$S = (2, -2, 2)$

$$\vec{a} = \vec{P}\vec{Q} = (4-2, 1-0, 0-(-1)) = (2, 1, 1)$$

$$\vec{b} = \vec{P}\vec{R} = (3-2, -1-0, 1-(-1)) = (1, -1, 2)$$

$$\vec{c} = \vec{P}\vec{S} = (2-2, -2-0, 2-(-1)) = (0, -2, 3)$$

$$\vec{a} \cdot (\vec{b} \times \vec{c}) = \begin{vmatrix} 2 & 1 & 1 \\ 1 & -1 & 2 \\ 0 & -2 & 3 \end{vmatrix} = 2 \begin{vmatrix} -1 & 2 \\ -2 & 3 \end{vmatrix} - 1 \begin{vmatrix} 1 & 2 \\ 0 & 3 \end{vmatrix} + 1 \begin{vmatrix} 1 & -1 \\ 0 & -2 \end{vmatrix} = -3$$

The volume of the parallelepiped = $|\vec{a} \cdot (\vec{b} \times \vec{c})| = |-3| = 3$

13 Use scalar triple product to verify that the

vectors $\vec{a} = 2\vec{i} + 3\vec{j} + \vec{k}$ $\vec{b} = \vec{i} - \vec{j}$ $\vec{c} = 7\vec{i} + 3\vec{j} + 2\vec{k}$

$$\vec{a} = \langle 2, 3, 1 \rangle \quad \vec{b} = \langle 1, -1, 0 \rangle \quad \vec{c} = \langle 7, 3, 2 \rangle$$

$$\vec{a} \cdot (\vec{b} \times \vec{c}) = \begin{vmatrix} 2 & 3 & 1 \\ 1 & -1 & 0 \\ 7 & 3 & 2 \end{vmatrix} = 2 \begin{vmatrix} -1 & 0 \\ 3 & 2 \end{vmatrix} - 3 \begin{vmatrix} 1 & 0 \\ 7 & 2 \end{vmatrix} + 1 \begin{vmatrix} 1 & -1 \\ 7 & 3 \end{vmatrix} = -4 - 6 + 10 = 0$$

[Note Properties of determinant. If two rows of A are interchanged to produce B, then $\det B = -\det A$. So change the rows of $\vec{a} \cdot (\vec{b} \times \vec{c})$ will give zero.]

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(14) Find the line through the point $(1, 0, -3)$ and parallel to the vector

$$2\vec{i} - 4\vec{j} + 5\vec{k}$$

$$\vec{r} = \vec{r}_0 + t\vec{v}$$

$$\vec{r}_0 = 1\vec{i} + 0\vec{j} - 3\vec{k}$$

$$\vec{v} = 2\vec{i} - 4\vec{j} + 5\vec{k}$$

$$\begin{aligned} (x, y, z) &= 1\vec{i} - 3\vec{k} + 2t\vec{i} - 4t\vec{j} + 5t\vec{k} \\ &= 3t\vec{i} + 2t\vec{k} - 4t\vec{j} \end{aligned}$$

(15) Find the line through the point $(1, 0, 6)$ and perpendicular

to the plane $x + 3y + z = 5$

$$P_0 = (1, 0, 6) \quad \vec{r}_0 = \langle 1, 0, 6 \rangle$$

$\rightarrow \vec{n}$ vector of the plane $x + 3y + z = 5$ is $\vec{n} = \langle 1, 3, 1 \rangle$

This vector is perpendicular to the plane.

$\rightarrow$ For line perpendicular to the plane we must have

$$\vec{r} = \vec{r}_0 + t\vec{v} \quad \text{here } \vec{v} \parallel \vec{n} \quad \text{so we must obtain the}$$

$$\text{equation of the line } \langle x, y, z \rangle = \langle 1, 0, 6 \rangle + t\langle 1, 3, 1 \rangle$$

$$\langle x, y, z \rangle = \langle 1+t, 3t, 6+t \rangle$$

(16) Find parametric and symmetric equations for the line through $(2, 1, 0)$ and perpendicular to both $\vec{i} + \vec{j}$ and $\vec{j} + \vec{k}$

$$\text{Let } \vec{u} = \vec{i} + \vec{j} = \langle 1, 1, 0 \rangle \quad \text{Let get } \vec{w} = \vec{u} \times \vec{v} \text{ such that}$$

$$\vec{v} = \vec{j} + \vec{k} = \langle 0, 1, 1 \rangle \quad \vec{w} \perp \vec{u}, \quad \vec{w} \perp \vec{v}$$

$$\vec{w} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix} = \vec{i}(1) - \vec{j}(1) + \vec{k}(1) = \langle 1, -1, 1 \rangle$$

$$\begin{aligned} \text{Equation of the line } \vec{r} &= \vec{r}_0 + t\vec{w} = \langle 2, 1, 0 \rangle + t\langle 1, -1, 1 \rangle \\ &= \langle 2+t, 1-t, t \rangle \end{aligned}$$

$$\text{Parametric } x = 2+t \quad y = 1-t \quad z = t$$

$$\text{Symmetric } \frac{x-2}{1} = \frac{y-1}{-1} = z$$

(9)

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- (17) Find equation of the line which is the intersection of the planes $x+y+z=1$ and $x+z=0$

$$x+y+z=1 \quad \text{and} \quad x+z=0 \quad x=-z$$

$$-z+y+z=1 \Rightarrow \text{Intersection equation} \quad \begin{matrix} x=t \\ y=1 \\ z=-t \end{matrix}$$

- (18) Is the line through $P=(-4, -6, 1)$ and $Q=(2, 0, -3)$ parallel to the line through $R=(10, 18, 4)$ and $T=(5, 3, 4)$

$$\text{The direction of the line } \vec{PQ} = \vec{v} = (-2 - (-4), 0 - (-6), -3 - 1) \\ \vec{v} = (2, 6, -4)$$

$$\text{The direction of the line } \vec{RT} = \vec{w} = (5 - 10, 3 - 18, 4 - 4) \\ \vec{w} = (-5, -15, 0) \quad \vec{w} = -\frac{5}{2} \cdot \vec{v} \quad \text{so they are parallel.}$$

- (19) Show that the lines L_1 and L_2 with parametric equations

$$\begin{matrix} x=1+t & y=-2+3t & z=4-t \\ x=2s & y=3+s & z=-3+4s \end{matrix}$$

are skew lines; that is, they do not intersect and not parallel.

→ The lines are not parallel because the vectors

$$\vec{v}_1 = \langle 1, 3, -1 \rangle$$

$$x=1+1t \quad y=-2+3t \quad z=4-1t$$

$$\vec{v}_2 = \langle 2, 1, 4 \rangle$$

$\vec{v}_1$ and $\vec{v}_2$ are not parallel.

→ not intersect Consider they intersect. such that

$$1+t=2s$$

$$t=2s-1$$

$$-2+3t=3+s$$

$$3t=5+s$$

$$4-t=-3+4s$$

$$t=7-4s$$

(10)

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Let's solve.
$$\begin{cases} t = 2s - 1 \\ 3t = 5 + s \\ t = 7 - 4s \end{cases} \quad \begin{cases} 3(2s - 1) = 5 + s \\ 6s - 3 = 5 + s \\ 5s = 8 \\ s = \frac{8}{5} \Rightarrow t = 2 \cdot \frac{8}{5} - 1 = \frac{11}{5} \end{cases}$$

But $s = \frac{8}{5}$ and $t = \frac{11}{5}$ don't satisfy the third equation.

$$t = 7 - 4s$$

$$\frac{11}{5} \neq 7 - 4 \cdot \frac{8}{5} \quad \frac{11}{5} \neq \frac{3}{5}$$

They do not intersect. NOT INTERSECT, NOT PARALLEL
SO SKEW.

(20) Determine whether the lines L_1 and L_2 are parallel, skew or intersecting. If they intersect find the point of intersection.

$$L_1: \frac{x-1}{1} = \frac{y-1}{2} = \frac{z-2}{3}$$

$$L_2: \frac{x-3}{-4} = \frac{y-2}{-3} = \frac{z-1}{2}$$

$$\vec{v}_1 \text{ for } L_1: \vec{v}_1 = \langle 1, 2, 3 \rangle$$

$$\vec{v}_2 = \langle -4, -3, 2 \rangle$$

Since $\vec{v}_1$ not a multiple of $\vec{v}_2$, they are not parallel

Intersect Write parametric equations

$$\begin{aligned} L_1: \quad x &= 1 + t \Rightarrow x = t \\ y - 1 &= 2t \Rightarrow y = 2t + 1 \\ z - 2 &= 3t \Rightarrow z = 3t + 2 \end{aligned}$$

$$\begin{aligned} L_2: \quad x - 3 &= -4s \Rightarrow x = 3 - 4s \\ y - 2 &= -3s \Rightarrow y = 2 - 3s \\ z - 1 &= 2s \Rightarrow z = 2s + 1 \end{aligned}$$

$$\begin{cases} t = 3 - 4s & t + 4s = 3 \\ 2s + 1 = 3t + 2 & 3t - 2s = -1 \end{cases} \Rightarrow \begin{cases} 7t = 1 \\ t = 1/7 \\ s = 5/7 \end{cases}$$

Let's see if $t = 1/7, s = 5/7$ satisfies.

$$y = 2t + 1 \quad \text{and} \quad y = 2 - 3s$$

$$y = \frac{2}{7} + 1 \neq 2 - \frac{15}{7} \quad \text{NOT INTERSECT}$$

NOT INTERSECT, NOT PARALLEL SO SKEW

(11)

Daya Altıntaş

21 Find the equation of the plane through the point $(-1, 6, -5)$ and parallel to the plane $x + y + z + 2 = 0$

Normal of $x + y + z + 2 = 0$ is $\vec{n}_1 = \langle 1, 1, 1 \rangle$

Since we want to obtain the plane parallel to that line, this means that normals will be parallel.

If $\vec{n} = (a, b, c)$ and plane passes through (x_0, y_0, z_0)
equation of the plane $\Rightarrow a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$

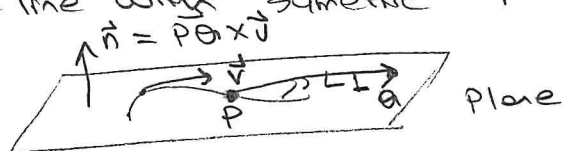
Normal of the plane $\vec{n}_2 = \alpha \vec{n}_1 = \langle \alpha, \alpha, \alpha \rangle$

$$\alpha(x + 1) + \alpha(y - 6) + \alpha(z + 5) = 0$$

$$\underline{(x + 1) + (y - 6) + (z + 5) = 0}$$

21 Find equation of the plane that passes through the point $(1, -1, 1)$ and contains the line with symmetric equation.

$$L: x = 2y = 3z$$



$$L: \frac{x}{1} = \frac{y}{1/2} = \frac{z}{1/3} \quad \text{its direction vector } \vec{v} = \langle 1, 1/2, 1/3 \rangle$$

Thus, find a point on that line (choose any point)

$(0, 0, 0)$ is on the line. But $(1, -1, 1)$ is not on the line but on the plane.

Now we have two points $P = (0, 0, 0)$ $\theta = (1, -1, 1)$
on line on plane

$\vec{P}\theta = \langle 1, -1, 1 \rangle$ is parallel to plane but not parallel to the line

$$\vec{P}\theta \times \vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 1/2 & 1/3 \\ 1 & -1 & 1 \end{vmatrix} = \langle 5/6, -2/3, -3/2 \rangle \text{ normal to the plane}$$

Equation of the plane $\frac{5}{6}(x - 0) - \frac{2}{3}(y - 0) - \frac{3}{2}(z - 0) = 0$

Deep Attnley

(23) (a) Find the angle between the planes $x+y+z=1$ and $x-2y+3z=1$

(b) find symmetric equations for line of intersection L of these two planes.

(c) $x+y+z=1 \Rightarrow \text{normal } \vec{n}_1 = \langle 1, 1, 1 \rangle$

$x-2y+3z=1 \Rightarrow \text{normal } \vec{n}_2 = \langle 1, -2, 3 \rangle$

Angle between planes = angle between normals.

$$\vec{n}_1 \cdot \vec{n}_2 = |\vec{n}_1| \cdot |\vec{n}_2| \cdot \cos \theta$$

$$\cos \theta = \frac{1 \cdot 1 + 1(-2) + 1 \cdot 3}{\sqrt{1+1+1} \sqrt{1+4+9}} = \frac{2}{\sqrt{3} \sqrt{14}} \quad \theta = \arccos\left(\frac{2}{\sqrt{42}}\right) = 72^\circ$$

(b) At first we need a point on intersection line L .

We can find the point of intersection xy plane set $z=0$

(You can use also $x=0$ or $y=0$ depends on you)

$$x+y+0=1$$

$$x+y=1$$

$$\left. \begin{array}{l} x+y=1 \\ x-2y=1 \end{array} \right\} \begin{array}{l} x=1 \quad y=0 \quad z=0 \\ (1,0,0) \text{ point is on the line } L \end{array}$$

$$x-2y+3 \cdot 0=1$$

$$x-2y=1$$

(1,0,0) point is on the line L .

Now, let's find its direction vector.

Since L lies in both planes this means that it is perpendicular to both normals $\vec{n}_1$ and $\vec{n}_2$.

$$\vec{v} = \vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 1 & 1 \\ 1 & -2 & 3 \end{vmatrix} = 5\vec{i} - 2\vec{j} - 3\vec{k} = \langle 5, -2, -3 \rangle$$

equation of line is $\langle x, y, z \rangle = \langle 1, 0, 0 \rangle + t \langle 5, -2, -3 \rangle$

$$x = 1 + 5t \quad y = -2t \quad z = -3t$$

Deyan Atanasov

(24) Find equation of the plane that passes through the point $(-1, 2, 1)$ and contains the line of intersection of the planes $x+y-z=2$ and $2x-y+3z=1$.

Consider L is the intersection line, normal of $P_1: x+y-z=2$ is $\vec{n}_1$ and $P_2: 2x-y+3z=1$ is $\vec{n}_2$.

Since L is in the intersection this means that it is on plane P_1 and therefore perpendicular to $\vec{n}_1$, similarly perpendicular to $\vec{n}_2$. So, it is parallel to $\vec{n}_1 \times \vec{n}_2$.

$$\vec{v} = \vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 1 & -1 \\ 2 & -1 & 3 \end{vmatrix} = \langle 2, -5, -3 \rangle$$

direction of L . The desired plane is parallel to $\vec{v}$.

Find a point at intersection choose $x=0$ $y-z=2$

$3z-y=1$, solve together

$$\begin{cases} y-z=2 \\ 3z-y=1 \end{cases} \Rightarrow z = \frac{3}{2} \quad y = \frac{7}{2} \quad P(0, 7/2, 3/2)$$

→ Another vector parallel to the desired plane is the vector $\vec{PQ}$ containing the points P and Q .

$$\vec{PQ} = \underbrace{P(0, 7/2, 3/2)}_{\text{point on intersection line}} - \underbrace{Q(-1, 2, 1)}_{\text{point on the plane}}$$

$$\vec{PQ} = \langle -1, -3/2, -1/2 \rangle$$

Now, normal equation of the plane that we want to obtain is cross product of parallel vectors to the plane.

$$\vec{N} = \vec{v} \times \vec{PQ} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & -5 & -3 \\ -1 & -3/2 & -1/2 \end{vmatrix} = \langle -2, 4, -8 \rangle$$

$$\text{Equation of the plane } -2(x - (-1)) + 4(y - 2) - 8(z - 1) = 0$$

Deyi Altinler

- 25 Find the point at which the line $x=3-t$, $y=2+t$, $z=5t$ intersects the plane $x-y+2z=9$

Insert equation of line to the plane $\begin{matrix} x-y+2z=9 \\ \downarrow \quad \quad \quad \uparrow \\ 3-t-(2+t)+10t=9 \end{matrix}$

$$3-t-2-t+10t=9 \quad 1+8t=9 \quad t=1 \text{ point is}$$

$$(x=3-t \Rightarrow x=2 \quad (2, 3, 5))$$

$$y=2+t \Rightarrow y=3$$

$$z=5t \Rightarrow z=5$$

- 26 Where does the line through (1, 0, 1) and (4, -2, 2) intersect the plane $x+y+z=6$.

Direction vector $\vec{v}$ of the line is $\vec{v} = \vec{PQ} = (4, -2, 2) - (1, 0, 1)$

$$\vec{v} = (3, -2, 1)$$

Equation of the line can be obtained as $\frac{x-1}{3} = \frac{y-0}{-2} = \frac{z-1}{1}$

$$x = 1+3t \quad z = t+1$$

$$y = -2t$$

Insert it to the plane $x+y+z=6 \Rightarrow 1+3t-2t+t+1=6$

$$2+2t=6 \quad t=2 \quad \text{so the point is}$$

$$(x, y, z) = (7, -4, 3)$$

- 28 Find symmetric equations for the line of intersection of the planes $\underbrace{x+y-z=2}_{\text{normal denoted } \vec{n}_1}$ and $\underbrace{3x-4y+5z=6}_{\text{normal denoted } \vec{n}_2}$

(b) find the angle between planes

(a) Line of intersection, Since line is in the intersection it must be perpendicular to both $\vec{n}_1$ and $\vec{n}_2$, so it is parallel to $\vec{n}_1 \times \vec{n}_2$

$$\vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 4 & -1 \\ 3 & -4 & 5 \end{vmatrix} = \vec{i}(5-4) - \vec{j}(5+3) + \vec{k}(-4-3) \\ = \vec{i} - 8\vec{j} - 7\vec{k} = \langle 1, -8, -7 \rangle$$

Find a point on line intersection of the planes $x+y+z=2$.

$$3x+4y+5z=6$$

Choose $y=0$.

$$3x+5z=6$$

$$x=2$$

Intersection point

$$x-z=2$$

$$z=0$$

$$(2, 0, 0)$$

Equation of line

$$x = 2 + t$$

$$y = 0 - 8t$$

$$z = 0 - 7t$$

or you can write

$$x-2 = \frac{y}{-8} = \frac{z}{-7}$$

(b) angle between planes = angle between normals

$$\vec{n}_1 = \langle 1, 1, -1 \rangle$$

$$\vec{n}_2 = \langle 3, -4, 5 \rangle$$

$$|\vec{n}_1| = \sqrt{3}$$

$$\vec{n}_1 \cdot \vec{n}_2 = 3 \cdot 1 - 4 \cdot 1 - 1 \cdot 5 = -6$$

$$|\vec{n}_2| = \sqrt{9+16+25} = \sqrt{50}$$

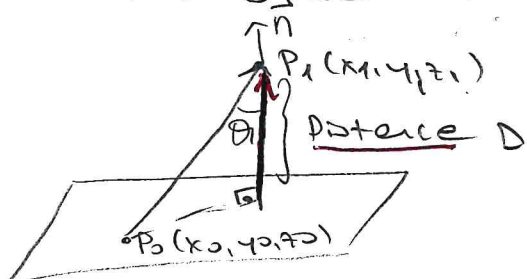
$$\vec{n}_1 \cdot \vec{n}_2 = |\vec{n}_1| \cdot |\vec{n}_2| \cdot \cos \theta \quad \cos \theta = \frac{-6}{\sqrt{3} \cdot \sqrt{50}} \approx 11.9^\circ$$



(29) (a) Find the formula for the distance D from the point $P_1(x_1, y_1, z_1)$ to the plane $ax+by+cz+d=0$

(b) Using that formula find the distance between $(1/2, 0, 0)$ and the plane $5x+y+z-1=0$.

(c) Let $P_0(x_0, y_0, z_0)$ be any point on the plane and let $\vec{b}$ be the $\vec{P_0P_1}$. Then, $\vec{b} = \langle x_1-x_0, y_1-y_0, z_1-z_0 \rangle$



The distance D is equal to the absolute value of the scalar projection of $\vec{b}$ to the normal $\vec{n}$

Thus,

$$D = |\text{comp}_{\vec{a}} \vec{b}| = \left| \frac{\vec{n} \cdot \vec{b}}{|\vec{b}|} \right| = \frac{|ax_1 + by_1 + cz_1 - \overbrace{(ax_0 + by_0 + cz_0)}^{-d}|}{\sqrt{a^2 + b^2 + c^2}}$$

Then, since P_0 satisfies equation of the plane this means that $ax_0 + by_0 + cz_0 + d = 0$. Then D can be written as

$$D = \frac{|ax_1 + by_1 + cz_1 + d|}{\sqrt{a^2 + b^2 + c^2}}$$

(b) Point $(1/2, 0, 0)$, Plane $5x + y - z = 1 \Rightarrow$

$$D = \frac{|5 \cdot 1/2 + 0 - 0 - 1|}{\sqrt{5^2 + 1 + 1}} = \frac{3/2}{\sqrt{27}} = \frac{\sqrt{3}}{6}$$

(30) Show that the distance between the parallel lines $ax + by + cz + d_1 = 0$ and $ax + by + cz + d_2 = 0$ is

$$D = \frac{|d_1 - d_2|}{\sqrt{a^2 + b^2 + c^2}}$$

Consider $P_0 = (x_0, y_0, z_0)$ is on the plane $ax + by + cz + d_1 = 0$. Then, using the distance formula between a point P_0 and the plane $ax + by + cz + d_2 = 0$ is

$$D = \frac{|ax_0 + by_0 + cz_0 + \overbrace{d_2}^{-d_1}|}{\sqrt{a^2 + b^2 + c^2}} = \frac{|d_1 - d_2|}{\sqrt{a^2 + b^2 + c^2}}$$

31 Find the domain of the vector function

a) $\vec{r}(t) = \langle t^2, \sqrt{t-1}, \sqrt{5-t} \rangle$

b) $\vec{r}(t) = \frac{t-2}{t+2} \vec{i} + \sin t \vec{j} + \ln(9-t^2) \vec{k}$

a) $t^2 \Rightarrow D_f = (-\infty, \infty)$

$D_{\vec{r}}(t) = D_f \cap D_g \cap D_h$
 $= [1, 5]$

$g(t) = \sqrt{t-1} \Rightarrow D_g = [1, \infty) \quad t-1 \geq 0 \quad t \geq 1$

$h(t) = \sqrt{5-t} \Rightarrow D_h = (-\infty, 5] \quad 5-t \geq 0 \quad t \leq 5$

b) $f(t) = \frac{t-2}{t+2} \quad D_f = \mathbb{R} \setminus \{-2\}$

$D_{\vec{r}}(t) = (-3, -2) \cup (-2, 3)$

$g(t) = \sin t \quad D_g = \mathbb{R}$

$h(t) = \ln(9-t^2) \quad D_h = (-3, 3) \quad 9-t^2 > 0 \quad t^2 < 9 \quad -3 < t < 3$

32 Evaluate the following limits.

a) $\lim_{t \rightarrow 1} \sqrt{t+3} \vec{i} + \frac{t-1}{t^2-1} \vec{j} + \frac{t \cos t}{t} \vec{k} = ?$

$\lim_{t \rightarrow 1} \sqrt{t+3} = 2$

$\lim_{t \rightarrow 1} \frac{t-1}{t^2-1} = \lim_{t \rightarrow 1} \frac{t-1}{(t-1)(t+1)} = \frac{1}{2}$

$\lim_{t \rightarrow 1} \frac{t \cos t}{t} = \cos 1$

$\lim_{t \rightarrow 1} \sqrt{t+3} \vec{i} + \frac{t-1}{t^2-1} \vec{j} + \frac{t \cos t}{t} \vec{k} = 2\vec{i} + \frac{1}{2}\vec{j} + \cos 1 \vec{k}$

b) $\lim_{t \rightarrow \infty} \langle \arctan t, e^{-2t}, \frac{\ln t}{t} \rangle = ?$

$\lim_{t \rightarrow \infty} \arctan t = \pi/2$

$\lim_{t \rightarrow \infty} \frac{\ln t}{t} \stackrel{\text{CH}}{=} \lim_{t \rightarrow \infty} \frac{1/t}{1} = 0$

$\lim_{t \rightarrow \infty} e^{-2t} = 0 \quad \lim_{t \rightarrow \infty} \langle \arctan t, e^{-2t}, \frac{\ln t}{t} \rangle = \langle \pi/2, 0, 0 \rangle$

- (33) Find a vector equation and parametric equations for the line segment joining P and Q $P = (1, 0, 1)$ $Q = (2, 3, 1)$

Parametric equation of joining two points

$$\vec{r}(t) = (1-t)\vec{r}_0 + t\vec{r}_1 \quad 0 \leq t \leq 1$$

$\uparrow$ Position vector of P $\uparrow$ Position vector of Q

$$\begin{aligned} \vec{r}(t) &= (1-t)\langle 1, 0, 1 \rangle + t\langle 2, 3, 1 \rangle \\ &= \langle (1-t) + 2t, 3t + (1-t)t, t \rangle \\ &= \langle 1+t, 3t, 1 \rangle \quad 0 \leq t \leq 1 \end{aligned}$$

- (34) Find the unit tangent vector $\vec{T}(t)$ at the point with the given value of the parameter t .

a) $\vec{r}(t) = \langle 6t^5, 4t^3, 2t \rangle \quad t=1$

b) $\vec{r}(t) = \cos t \vec{i} + 3t \frac{\vec{j}}{5} + 2 \sin 2t \vec{k} \quad t=0$

c) $\vec{r}'(t) = \langle 30t^4, 12t^2, 2 \rangle \quad \vec{T}(1) = \frac{\vec{r}'(1)}{|\vec{r}'(1)|} = \frac{\langle 30, 12, 2 \rangle}{\sqrt{(30)^2 + (12)^2 + 4^2}}$

d) $\vec{r}'(t) = -\sin t \vec{i} + 3 \frac{\vec{j}}{5} + 4 \cos 2t \vec{k} \quad \vec{r}'(0) = 0\vec{i} + 3\frac{\vec{j}}{5} + 4\vec{k}$
 $\vec{T}(0) = \frac{\vec{r}'(0)}{|\vec{r}'(0)|} = \frac{\langle 0, 3, 4 \rangle}{\sqrt{9+16}} = \langle 0, \frac{3}{5}, \frac{4}{5} \rangle$

- (35) Find parametric equation for the tangent line to the curve $x=t^5, y=t^4, z=t^3 \quad (1, 1, 1)$

$$\vec{r}(t) = \langle t^5, t^4, t^3 \rangle \quad \vec{r}'(t) = \langle 5t^4, 4t^3, 3t^2 \rangle$$

The tangent vector $\vec{r}'(1) = \langle 5, 4, 3 \rangle$

Tangent line = The line passing through $(1, 1, 1)$ and parallel to $\vec{r}'(1)$
 $\langle x, y, z \rangle = \langle 1, 1, 1 \rangle + t \langle 5, 4, 3 \rangle$
 $x = 1 + 5t$
 $y = 1 + 4t$
 $z = 1 + 3t$

3.6 At what points do the curves $r_1(t) = \langle t, 1-t, 3+t^2 \rangle$ and $r_2(s) = \langle 3-s, s-2, s^2 \rangle$ intersect? Find their angle of intersection.

Solve $r_1(t) = r_2(s)$

$$\begin{aligned} (1) \quad t &= 3-s \\ (2) \quad 1-t &= s-2 \\ (3) \quad s^2 &= 3+t^2 \end{aligned} \quad \left. \begin{array}{l} t=3-s \\ 1-(3-s)=s-2 \\ -2+s=s-2 \end{array} \right\} \begin{array}{l} t=3-s \\ \text{we couldn't} \\ \text{obtain anything} \\ \text{by solving eq (1)} \\ \text{and (2).} \end{array}$$

let's try to solve $1-t=s-2$
 $s^2 = 3+t^2$

$$1-t=s-2 \Rightarrow t=3-s \quad s^2 = 3+t^2 \Rightarrow s^2 = 3+(3-s)^2$$

$$s^2 = 3 + (9 - 6s + s^2) \quad 6s = 12 \quad \underline{s=2 \quad t=1}$$

Check if it satisfies Eq (1), if yes there is an intersection.

$$t = 3-s \quad 1 = 3-2 \Rightarrow 1=1 \quad \checkmark$$

So, the intersection point $s=2 \quad (1, 0, 4)$

To find the angle of intersection, we will look to the tangent vectors at that point

$$r_1'(t) = \langle 1, -1, 2t \rangle \quad r_1'(t) \Big|_{t=1} = \langle 1, -1, 2 \rangle \quad |r_1'(1)| = \sqrt{6}$$

$$r_2'(s) = \langle -1, 1, 2s \rangle \quad r_2'(s) \Big|_{s=2} = \langle -1, 1, 4 \rangle \quad |r_2'(2)| = \sqrt{18}$$

angle between $r_1'(t)$ and $r_2'(s)$

$$r_1'(t) \cdot r_2'(s) = |r_1'(t)| |r_2'(s)| \cos \theta$$

$$\cos \theta = \frac{-1 - 1 + 8}{\sqrt{6} \cdot \sqrt{18}} = \frac{1}{\sqrt{3}}$$

$$\theta = \arccos\left(\frac{1}{\sqrt{3}}\right) \approx 55^\circ$$

Do you Altinter

37 Determine if the curve $\vec{r}(t) = \langle t^3, t^4, t^5 \rangle$ is smooth?

$$\vec{r}(t) = \langle t^3, t^4, t^5 \rangle \quad \vec{r}'(t) = \langle 3t^2, 4t^3, 5t^4 \rangle \quad \text{Both.}$$

Smooth curve $\vec{r}(t)$ and $\vec{r}'(t)$ is continuous and $\vec{r}'(t) \neq 0$ except possibly at the endpoints

$\vec{r}(t)$ and $\vec{r}'(t)$ are both continuous. But, $\vec{r}'(0) = \langle 0, 0, 0 \rangle$ so, it is not smooth.

38 Find the lengths of the following curves

a) $\vec{r}(t) = \langle 2 \sin t, 5t, 2 \cos t \rangle \quad -10 \leq t \leq 10$

b) $\vec{r}(t) = t\vec{i} + 2t\vec{j} + \ln t\vec{k} \quad 1 \leq t \leq e$

a) $L = \int_a^b \sqrt{(x'(t))^2 + (y'(t))^2 + (z'(t))^2} dt = \int_a^b |\vec{r}'(t)| dt.$

Length of $\vec{r}(t)$
between $a \leq t \leq b$

$$L = \int_{-10}^{10} \sqrt{4 \cos^2 t + 25 + 4 \sin^2 t} dt \\ = \int_{-10}^{10} \sqrt{29} dt = 20\sqrt{29}$$

$$x'(t) = 2 \cos t$$

$$y'(t) = 5$$

$$z'(t) = -2 \sin t$$

b) $\vec{r}'(t) = 2t\vec{i} + 2\vec{j} + \frac{1}{t}\vec{k} \quad |\vec{r}'(t)| = \sqrt{4t^2 + 4 + \frac{1}{t^2}}$

$$= \frac{1+2t^2}{t}$$

since $1 \leq t \leq e$ we have $|\vec{r}'(t)| = \frac{1+2t^2}{t}$

$$\int_1^e \frac{1+2t^2}{t} dt = \ln t + 2t^2 \Big|_1^e = e^2.$$

Deep Atkinson

(39) Find unit tangent vectors $T(t)$ and $N(t)$ for

$$r(t) = \langle \sqrt{2}t, e^t, e^{-t} \rangle$$

$$\text{Tangent } T(t) = \frac{r'(t)}{|r'(t)|} = \frac{\langle \sqrt{2}, e^t, -e^{-t} \rangle}{\sqrt{(\sqrt{2})^2 + (e^t)^2 + (-e^{-t})^2}}$$

$$|r'(t)|$$

$$\sqrt{(\sqrt{2})^2 + (e^t)^2 + (-e^{-t})^2}$$

$$= \frac{\langle \sqrt{2}, e^t, -e^{-t} \rangle}{\sqrt{2 + e^{2t} + e^{-2t}}} = \frac{\langle \sqrt{2}, e^t, -e^{-t} \rangle}{\sqrt{(e^t + e^{-t})^2}}$$

$$\sqrt{2 + e^{2t} + e^{-2t}}$$

$$\sqrt{(e^t + e^{-t})^2}$$

$$= \frac{\langle \sqrt{2}, e^t, -e^{-t} \rangle}{e^t + e^{-t}} = \left\langle \frac{\sqrt{2}}{e^t + e^{-t}}, \frac{e^t}{e^t + e^{-t}}, -\frac{e^{-t}}{e^t + e^{-t}} \right\rangle$$

Normal $N(t)$

$$N(t) = \frac{T'(t)}{|T'(t)|} \text{ How?}$$

$$|T'(t)|$$

$$|T(t)| = 1$$

$$\text{So, } T(t) \cdot T'(t) = 0$$

which means that

$$T(t) \perp T'(t) \text{ So unit normal } N(t) = \frac{T'(t)}{|T'(t)|}$$

$$T'(t) = \frac{1}{e^{2t}+1} \langle \sqrt{2}e^t, 2e^t, 0 \rangle - \frac{2e^{2t}}{(e^{2t}+1)^2} \langle \sqrt{2}e^t, e^{2t}, -1 \rangle$$

$$= \frac{1}{(e^{2t}+1)^2} \langle \sqrt{2}e^t(1-e^{2t}), 2e^{2t}, 2e^{2t} \rangle$$

$$|T'(t)| = \frac{\sqrt{(\sqrt{2}e^t(1-e^{2t}))^2 + (2e^{2t})^2 + (2e^{2t})^2}}{(e^{2t}+1)^4} = \frac{\sqrt{2}e^t}{e^{2t}+1}$$

$$N(t) = \frac{T'(t)}{|T'(t)|} = \frac{\langle 1-e^{2t}, \sqrt{2}e^t, \sqrt{2}e^t \rangle}{e^{2t}+1}$$

(40) Evaluate the integral. $I = \int_0^1 (\sqrt{t} \vec{i} + t e^{-t} \vec{j} + t^{-2} \vec{k}) dt = ?$

$$I = \vec{i} \cdot \frac{t^{3/2}}{3/2} \Big|_0^1 + \vec{j} \cdot \left(-t \cdot e^{-t} - e^{-t} \right) \Big|_0^1 + \vec{k} \cdot \frac{t^{-1}}{-1} \Big|_0^1 = \frac{2\sqrt{2}}{3} \vec{i} + \left(\frac{2-5e^{-3}}{e} \right) \vec{j} + \frac{3}{4} \vec{k}$$

$$\int u \cdot dv = u \cdot v - \int v \cdot du$$

$$\int t \cdot e^{-t} dt = -t \cdot e^{-t} + \int e^{-t} dt = -t \cdot e^{-t} - e^{-t} + C$$

$$t=u \quad e^{-t} dt = dv$$

$$dt = du \quad -e^{-t} = v$$

(22)

Deyu Altintan